

CHAPTER 1: INTRODUCTION

Communication is a fundamental part of human interaction, yet millions of people who are deaf or hard of hearing face daily challenges in expressing themselves and understanding others. Indian Sign Language (ISL) is the primary mode of communication for the hearing-impaired community in India, but widespread awareness and interpretation support remain limited. As a result, real-time communication between hearing and non-hearing individuals often requires human interpreters, which is neither scalable nor accessible in many situations.

Samvaad is an AI-powered translation system designed to bridge this communication gap by converting sign language into text/speech and text/speech into sign language in real time. The system integrates modern computer vision techniques, deep learning models, and natural language processing to interpret hand gestures captured through a webcam or images. It identifies hand landmarks using MediaPipe Hands, extracts features, and classifies gestures using a trained machine learning model. It also offers reverse translation using stored ISL templates, enabling users to learn or understand sign language visually.

The platform provides a seamless, user-friendly interface built with Streamlit, offering multiple modules such as Sign-to-Text, Text-to-Sign, Speech-to-Sign, Practice Mode, and personalized Analytics. These features make Samvaad not only a translation tool but also an educational and assistive system that promotes inclusivity, accessibility, and learning of Indian Sign Language.

By combining machine learning, computer vision, and speech processing, Samvaad aims to create a meaningful and scalable technological solution to support the deaf and mute community and encourage wider understanding of ISL among the general population.

1.1 Background

Communication is a fundamental aspect of human interaction, enabling people to express ideas, emotions, and information. However, individuals with speech and hearing impairments often face significant barriers when interacting with those who do not understand sign language. In India, the Indian Sign Language (ISL) is widely used within the Deaf community, yet very few people outside this community are familiar with it. This communication gap can lead to misunderstandings, limited accessibility, and social exclusion.

With the rapid growth of Artificial Intelligence, machine learning, and computer vision technologies, new possibilities have emerged to bridge this communication divide. Systems that can recognise hand gestures, interpret sign language, and convert it into spoken or written text are becoming increasingly feasible. At the same time, tools that allow text or speech to be converted into visual signs can support learning, accessibility, and inclusiveness for both hearing-impaired and non-signing individuals.

The **Samvaad** project was developed to address this need by creating an intelligent, interactive communication platform capable of translating between sign language, text, and speech. It uses MediaPipe for real-time hand landmark detection, a custom trained deep learning model for gesture classification, and Streamlit for an accessible, user-friendly interface. The system supports multiple translation modes—Sign-to-Text, Text-to-Sign, Speech-to-Sign, and Sign-to-Speech—making it useful for learning, conversation assistance, and inclusive communication.

As digital accessibility becomes an essential part of modern technology, solutions like Samvaad can play a transformative role in empowering speech- and hearing-impaired individuals, promoting inclusivity, and reducing communication barriers in everyday interactions.

1.2 Literature Survey

Sl. No	Publication	Year	Methodology	Result	Remarks
1	Zeyu Liang, Huailing Li, Jianping Chai, "Sign Language Translation: A Survey of Approaches and Techniques," Electronics, vol. 12, no. 12, p. 2678, 2023	2023	Survey and classification of existing methods for sign language translation (SLT) including improving sign language recognition (SLR), network improvements for translation accuracy, and addressing scarcity of sign language data. Reviewed encoder-decoder models, transformer-based architectures, multitask approaches, and data augmentation techniques.	Summarized state-of-the-art models and compiled datasets used in SLT research. Highlighted advantages of transformer-based end-to-end models and importance of multimodal input features like hand, face, and body posture.	Summarized state-of-the-art models and compiled datasets used in SLT research. Highlighted advantages of transformer-based end-to-end models and importance of multimodal input features like hand, face, and body posture.

2	M. Rajmohan, O. Ranga Babu, B. Sai Kumar Reddy, C. Srinivasan, S. Murugan, "Efficient Indian Sign Language Interpreter For Hearing Impaired," 2023 Second International Conference on Smart Technologies For Smart Nation (SmartTechCon), 2023	2023	Developed an end-to-end video-based ISL recognition system using image acquisition (camera/Kinect/Leap Motion), image preprocessing (histogram equalization, segmentation), feature extraction with Histogram of Oriented Gradients (HOG), and classification with Artificial Neural Networks (ANN). Data gathered includes 23 ISL signs with 98 images.	Achieved average accuracy of 92.33% on ISL detection; training accuracy of 94.98% and testing accuracy of 86.61% with HOG features and ANN classifier. Demonstrated effective static gesture recognition for deaf communication.	The system addresses static ISL gesture recognition; limitations include potential misclassification with similar gestures, small dataset size, and lack of continuous sign or facial expression recognition. Future directions include expanding datasets, enhancing accuracy with AI/ML, and incorporating VR/AR for immersive learning.
3	Muhammad Al-Qurishi, Thariq Khalid, Riad Souissi, "Deep Learning for Sign Language Recognition: Current Techniques, Benchmarks, and Open Issues," IEEE Access, 2021	2021	Comprehensive review of machine learning and deep learning techniques for sign language recognition (SLR), including CNN, RNN, LSTM, DBN, RCNN, hybrid models, and transformer-based approaches. Analyzed datasets, input modalities (vision-based, sensor-based), and evaluation metrics	Summarized state-of-the-art architectures and training results showing promising accuracy levels, especially with hybrid and transformer models. Highlighted that continuous SLR remains a challenge with accuracy generally lower than isolated recognition.	Identified key challenges such as dataset scarcity, regional sign language variations, continuous sign language complexity, multimodal data fusion, and real-time processing requirements. Suggested future work in building larger datasets, optimizing architectures, and improving generalization for real-world use.
4	Deep Kothadiya, Chintan Bhatt, Krenil Sapariya, Kevin Patel, Ana-Belén Gil-González, Juan M. Corchado, "Deepsign Sign	2022	Proposed a feedback-based deep learning model combining LSTM and GRU layers, using visual features extracted by InceptionResNetV2 from videos of 11 ISL gestures (IISL2020	LSTM-GRU model achieved ~97% accuracy in recognizing 11 isolated Indian sign language words; performance	Model works for isolated signs only (not continuous sentences); small vocabulary; no use of facial expressions; real-time, but limited by

	Language Detection and Recognition Using Deep Learning”.		dataset, recorded in natural conditions, ~1100 samples per sign, 16 participants). No extra sensors or gloves were used.	evaluated with K-fold cross-validation.	dataset size and environmental variability; recommends future work on larger datasets and advanced models (transformers).
5	Nikolas Adaloglou, Theocharis Chatzis, Ilias Papastratis, Andreas Stergioulas, Georgios Th. Papadopoulos, Vassia Zacharopoulou, George J. Xydopoulos, Klimnis Atzakas, Dimitris Papazachariou, Petros Daras, “A Comprehensive Study on Deep Learning-Based Methods for Sign Language Recognition”, IEEE Transactions on Multimedia, 2022	2022	Conducted a comparative experimental assessment of state-of-the-art deep learning SLR methods including 2D CNN-LSTM, 3D CNNs (I3D, 3D-ResNet), and hybrid models across multiple public datasets. Introduced new sequence training criteria (EnCTC, StimCTC) to improve gloss temporal alignment and modeling. Developed a large Greek Sign Language RGBD dataset for holistic evaluation.	Demonstrated that 3D CNN architectures excel at isolated sign recognition (~highest accuracy), while 2D CNN-LSTM models perform better in continuous sign language recognition. The new training criteria lowered error rates and improved temporal dependencies modeling.	Identified challenges in modeling non-manual features and subtle gloss boundaries. Limitations in handling continuous signing and coarticulation. Noted dataset size and diversity constraints. Suggested future work on multi-modal data integration and bridging recognition to translation.
6	Malay Kumar, S. Sarvajit Visagan, Tanish Mahajan, Anisha Natarajan, Sreeja P S, “Enhanced Sign Language Translation between American Sign Language and Indian Sign Language Using LLMs”, IEEE Access, 2025	2025	Proposed a hybrid deep learning model combining Random Forest Classifier and CNN for ASL gesture recognition; used a fine-tuned Large Language Model (LLM) for text correction; synthesized Indian Sign Language gestures from corrected text using RIFE-Net for smooth video generation; implemented an end-to-end translation pipeline from ASL to ISL.	Achieved 93% accuracy for ASL gesture recognition and 94.2% accuracy for text correction; real-time processing averaging 1 gesture per second and 60 FPS video output; successfully bridged semantic and cultural gap	Focused on static gesture recognition and translation; challenges remain in handling continuous gestures, dynamic movements, facial expressions, and larger vocabularies; framework currently limited to ASL and ISL with scope to extend to other languages; potential improvements include multi-

				between ASL and ISL.	modal inputs and adaptive learning.
7	Safi Ullah Chowdhury, Ahmed Shamir Shazid, Rifa Binte Rahmatullah, Nasima Begum, Tanjina Helaly, Rashik Rahman, "Descriptor Bengali Sign Word Recognition Video Dataset BdSLW102", IEEE Data, 2025	2025	Compiled a comprehensive Bengali sign word and sentence video dataset consisting of 102 word classes and 20 sentence classes with approximately 91,800 word videos and 6,000 sentence videos recorded under diverse real-world and controlled conditions. MediaPipe was used for hand, body, and facial landmark detection and extraction, generating raw and masked datasets. Privacy was ensured by applying Haar Cascade face blurring and encrypting original face data. The dataset supports machine learning model training with preprocessing standardized via keypoints extraction.	A deep learning baseline I3D model achieved 90.44% accuracy on unblurred videos and 77.83% accuracy on blurred videos, illustrating the impact of facial cues on sign recognition. The dataset is publicly available, enabling reproducible research and advancing Bengali sign language recognition.	The dataset captures environmental and signer variability and balances privacy with model accuracy through reversible face blurring. It facilitates word- and sentence-level Bengali sign recognition research, but recommends future work to incorporate facial features more comprehensively and explore privacy-accuracy trade-offs. This matches the style and level of detail of the paper 6 example you provided. Let me know if you want this in tabular form or with any modifications.
8	"A Portable System for Bidirectional Sign Language Translation for Deaf and Dumb," 2025	2025	Developed a bidirectional sign language translator using Convolutional Neural Networks for sign classification. System performs sign-to-text and text-to-sign conversions with a training accuracy of 94.86%.	Achieved 94.86% accuracy on training data; enables effective bidirectional communication between deaf/dumb and hearing individuals.	Provides an accessible communication interface facilitating social integration of hearing-impaired people; addresses misclassification issues with classification enhancement methods.
9	Lipisha Chaudhary et al., "SignNet II: A Transformer-Based	2022	Developed a transformer-based architecture jointly training sign-to-text and	Jointly trained networks outperformed	Advances two-way sign language translation,

SAMVAAD A Two-Way Real-Time Indian Sign Language Translator

	Two-Way Sign Language Translation Model," IEEE TPAMI, 2022		text-to-sign networks using a dual learning mechanism. Employed metric embedding learning to improve text-to-sign translation. Tested multi-feature transformers focusing on keypoint-based pose features.	single networks, with notable improvements in BLEU-1 to BLEU-4 scores on the German Sign Language dataset, showing up to 26% increase in BLEU-1.	bridging communication gaps. Pose features proved robust across video qualities. Model can continue to improve with more resources.
10	"Development of Sign Language Translator for Disable People in Two-ways Communication," 2025	2025	Developed a prototype using Python to translate hand gestures to text/speech and text to signs. Utilized camera tracking for recognizing hand motions, converted text input into GIF hand signs.	Achieved 99.3% accuracy. Enabled clear two-way communication between deaf/dumb and hearing people through an intuitive and free mobile-friendly system.	Python's character interpreter and open-source tools facilitate widespread accessibility; system offers potential to revolutionize hearing-impaired communication by bridging the gap effectively.
11	"Bidirectional Sign Language Assistant with MediaPipe Integration," 2023	2023	Developed a web-based system using MediaPipe for 3D pose estimation combined with machine learning for real-time sign recognition and translation between English and ASL. Features include video calls, animated avatars, and secure end-to-end communication.	Achieved high sign recognition accuracy and smooth 3D animation with real-time translation speed. User feedback highlighted usability and effectiveness.	System enables intuitive barrier-free communication between deaf and hearing users, with potential applications in education and personal communication. Future improvements include expanding dataset and adding contextual understanding.
12	Efficient learning for hearing-impaired: Two-Way Sign Language Translator using AI – Siddiqui, A.; Noor, A.; Saleem, M.	2024	AI/ML-based two-way translator using computer vision and NLP for real-time interaction	Achieved accurate bidirectional translation between sign gestures and text/speech	Focused on real-time applicability for hearing-impaired users in India

13	Real-Time Vision-Based Sign Language Bilateral Communication Device – Alam, M. R.; Ferdous, S. M.; et al	2023	Vision-based system using cameras and gesture recognition algorithms for bilateral communication	Successful prototype enabling real-time sign ↔ speech/text conversion	Demonstrates practical deployment; hardware-based device suitable for classroom/communication environments
14	A Real-Time Webcam-Based System for Sign Language and Speech Translation – Chowdhury, T.; Anwar, S.	2024	Webcam-based system combining gesture recognition and speech synthesis for two-way translation	Real-time translation achieved with moderate accuracy	Simple, cost-effective, suitable for personal/home use; emphasizes accessibility
15	Dhananjai Bajpai et al., “Two-Way American Sign Language Translator Glove System.”	2021	Developed a wearable ASL translator glove equipped with six flex sensors, three contact sensors, and an accelerometer, all linked to an AVR-2560 microcontroller. The glove captures hand gestures and sends recognized signs via Bluetooth, GSM/CDMA, or Internet for conversion into text, speech, or image outputs. A GUI enables gesture customization and response visualization.	Enabled creation of 1,300+ customizable gestures, achieving an average 83% translation accuracy in real-world scenarios. Provided portable bidirectional communication through wireless transmission.	Demonstrates effectiveness of sensor-based systems, but depends on hardware, lacks natural hand freedom, and is limited to ASL. Samvaad improves on this using vision-based, contactless, Indian Sign Language-focused translation.
16	Pulkit Singh et al., “A Python-Based Two-Way Sign Language Translator for Deaf and Dumb Communication.”	2022	Developed a two-way translator using CNN + Keras to convert real-time hand gestures to text/speech and text input to animated signs (GIFs). Used image acquisition, preprocessing, and vision-based gesture recognition for ISL letters and phrases. System applied letter-wise mapping and phrase matching to improve communication accuracy.	Achieved up to 99.3% accuracy on Indian Sign Language datasets in real-world testing. The system proved portable, fast, and user-friendly.	Shows a practical baseline for two-way ISL translation. However, relies on letter-level recognition rather than deep sequential models. Samvaad extends this by using MediaPipe + LSTM for dynamic gestures and real-time prediction.
17	Long Zhou et al., “Synchronous Bidirectional	2020	Proposed SB-NMT, a machine translation framework that generates	Achieved higher BLEU scores than strong	Although SB-NMT is text-based machine

	Neural Machine Translation (SB-NMT).”		target sentences simultaneously from left-to-right (L2R) and right-to-left (R2L) directions. Introduced synchronous bidirectional attention, enabling the model to leverage both past and future context during decoding. The system was evaluated on multiple language pairs including Chinese–English, English–German, and Russian–English.	Transformer baselines while maintaining the same number of parameters. Demonstrated improved output fluency, translation balance, and decoding speed.	translation, its bidirectional decoding concept is relevant to SAMVAAD, which also requires two-way (sign-to-text and text-to-sign) translation. Highlights potential for future integration of bidirectional attention mechanisms in sign language translation.
18	Lance Fernandes et al., “Bidirectional Sign Language Translation Using CNN and Sensor-Based Inputs.”	2022	Initially experimented with a sensor-based glove using flex sensors for gesture-to-text conversion but found limited accuracy. The study then transitioned to a CNN-based vision model, trained on a combination of existing and newly collected ASL datasets. The system supported two-way communication by enabling sign → text → speech and speech → text → sign translation with sequenced gesture video output.	Achieved 99%+ accuracy for ASL alphabet recognition using CNNs and custom datasets. Showed strong adaptability for introducing new gestures and facial features through dataset expansion.	Demonstrates effectiveness of vision-based CNNs compared to hardware gloves. Highlights the importance of diverse datasets and customizable gesture libraries. Relevant to SAMVAAD as it supports bidirectional translation , similar to the two-way ISL translation goal.
19	M. Suresh Anand et al., “Vision-Based Integrated Two-Way Indian Sign Language Translator.”	2021	Proposed a camera-based two-way ISL translation system requiring no gloves or sensors. The system operates in two modes: speech-to-sign (voice → text → matched signs → animated display) and sign-to-speech (gesture capture → preprocessing → segmentation → feature extraction → database matching → text/voice	Demonstrated an affordable and mobile-friendly solution for ISL translation. Showed effective recognition for alphabet and basic word-level gestures, emphasizing potential scalability for	Highlights an important vision-based baseline for Indian Sign Language. However, it relies heavily on rule-based matching and classical HMMs, whereas SAMVAAD advances this with MediaPipe, CNN/LSTM deep

			output). Techniques used include image binarization, edge detection, rule-based matching, noise filtering, and Hidden Markov Models (HMMs).	sentence-level ISL translation.	learning, and dynamic gesture recognition for better accuracy and real-time performance.
20	Prof. Shital Pawar et al., “Web-Based Bidirectional Real-Time Sign Language Assistant Using ML and MediaPipe.”	2023	Developed a web-based bidirectional ASL translation system supporting real-time sign-to-text and text-to-sign communication. Implemented MediaPipe for gesture tracking, 3D pose estimation, and animated avatars for sign visualization. Integrated CNNs and TensorFlow for precise gesture segmentation and recognition, and supported encrypted video calls for private communication.	Achieved high accuracy in recognizing ASL gestures and delivered interactive sign animations rated highly by both signers and non-signers. The system enabled smooth communication without a middle interpreter, improving accessibility for education and daily interactions.	Shows the capability of web-based ML systems for real-time sign translation. Samvaad extends this idea to Indian Sign Language, adds ISL animations, and improves contactless recognition using MediaPipe + LSTM with two-way translation.
21	Nada Shahin, Leila Ismail et al., “A Comprehensive Survey on Sign Language Machine Translation (SLMT).”	2023	Presented an in-depth review of SLMT evolution from rule-based and example-based systems to modern deep learning and transformer architectures. Discussed model types, datasets, gloss-to-text translation, isolated vs continuous sign recognition, and transformer designs (encoder-decoder, encoder-only, fusion).	Highlighted the superior flexibility and accuracy of transformer-based SLMT, and analyzed challenges such as data scarcity, privacy, and real-time constraints.	A strong theoretical foundation for two-way translation research. SAMVAAD builds on these insights by applying real-time ISL recognition using CNN/LSTM + MediaPipe and animation-based sign output.
22	M. Rajmohan et al., “Video-Based ISL Interpretation Using Camera-Based ANN and HOG Features.”	2021	Proposed a camera-based ISL recognition system using Kinect/Leap Motion. Applied segmentation, adaptive histogram equalization, and Histogram of Oriented Gradients (HOG) for feature	Achieved 94.98% training accuracy and 90.26% testing accuracy on 23 ISL signs. Demonstrated effectiveness of	Suitable for alphabet-level recognition but limited for dynamic signs and sentence translation. SAMVAAD extends this by using MediaPipe +

			extraction. Classification performed via Artificial Neural Networks (ANN).	HOG + ANN for static gestures.	LSTM for real-time dynamic gesture processing.
23	Diksha Kumari & Radhey Shyam Anand et al., “Hybrid Deep Learning for Video-Based Sign Language Recognition.”	2022	Introduced a hybrid model combining MobileNetV2 for spatial feature extraction and attention-based LSTM for temporal modeling using the WLASL dataset (100 word classes).	Achieved 84.65% average accuracy (5-fold cross-validation) and outperformed many CNN, RNN, and transformer baselines in efficiency.	Demonstrates high potential for lightweight real-time systems. SAMVAAD uses a similar sequential approach with LSTM, adapted for Indian Sign Language.
24	Subrata Kumer Paul et al., “Real-Time Sign Language Recognition Using CNN, LSTM, and GRU.”	2023	Built a real-time recognition system using enhanced ResNet-based CNN, LSTM, and GRU models. Collected two datasets: one with three sign phrases, another with 26 ASL alphabets using webcam/mobile devices.	Performance: LSTM – 94.3%, CNN – 89.07%, GRU – 79.3%. Validated real-time capability without sensors.	Confirms superiority of LSTM for temporal features. SAMVAAD uses LSTM for dynamic ISL gestures, aligning with this research.
25	Biao Zhang et al., “Scaling Sign Language Translation Using Large Multimodal Transformer Models.”	2024	Trained large-scale SLT models using 6,600+ hours of noisy YouTube sign videos, multilingual corpora, and synthetic data. Used mByT5 multi-task encoder-decoder transformers with multimodal and multilingual prompts.	Achieved state-of-the-art results and enabled zero-shot translation across 42 languages, showing strong generalization.	Shows how scale improves SLT accuracy. SAMVAAD is smaller but can upgrade toward such transformer-based large models in future enhancements.

1.3 Motivation

Communication is a basic human need, yet millions of people with hearing or speech impairments struggle to express themselves in environments where sign language is not commonly understood. In India, despite the presence of Indian Sign Language (ISL), awareness and usage among the general population remain low. This creates daily challenges for the Deaf community—whether in education, healthcare, workplaces, or simple day-to-day interactions.

The motivation behind developing **Samvaad** arises from the desire to bridge this communication gap using accessible and affordable technology. Instead of relying on expensive specialized equipment, this project leverages commonly available devices such as

laptops and webcams, combined with deep learning and computer vision, to create an intuitive and inclusive communication tool.

Another key motivation is the growing need for assistive technologies that support inclusive education and accessibility mandates. A system that allows Sign-to-Text, Text-to-Sign, Speech-to-Sign, and Sign-to-Speech translation can empower both hearing-impaired and non-signing individuals to interact more effectively. This project also aims to support learners who are beginning to explore ISL by offering visual, interactive tools to practice and improve their signing skills.

Beyond accessibility, the motivation also stems from the technological challenge and academic interest in gesture recognition, real-time processing, and multimodal AI integration. Building a system that can detect hand landmarks with precision, classify gestures accurately, and convert them into meaningful outputs provided an opportunity to practically apply machine learning, computer vision, and user-interface design concepts.

Overall, Samvaad is motivated by a blend of social impact, technological exploration, and the ambition to contribute toward a more inclusive digital ecosystem where communication barriers are reduced and opportunities for connection are expanded.

1.4 Problem Statement

Many people do not understand Indian Sign Language, making communication difficult for hearing and speech - impaired individuals. Current tools are limited and not real-time. A two-way ISL–English translator is needed to bridge this gap.

1.5 Aim and Objective

Aim

The primary aim of the *Samvaad* project is to develop an intelligent, real-time, and user-friendly communication system that bridges the interaction gap between the hearing-impaired community and non-signers by translating between **Indian Sign Language (ISL)** and spoken or written language using machine learning and computer vision techniques.

Objectives

1. To develop a vision-based recognition module Design and implement a computer vision system using OpenCV and deep learning models to accurately detect and classify Indian Sign Language (ISL) hand gestures in real time.
2. To enable two-way communication Create a translation framework that not only converts ISL gestures into English text/speech but also translates English text into animated ISL gestures for inclusive communication.
3. To integrate text-to-speech and speech-to-sign modules Implement a lightweight natural language processing layer that converts recognized ISL signs into audible English speech and converts English text input into corresponding ISL gesture animations.
4. To build a user-friendly interactive interface Develop a simple desktop or mobile interface that allows users to interact with the translator in real time, ensuring accessibility, responsiveness, and practical usability for hearing- and speech-impaired individuals.

1.6 Scope

The scope of the *Samvaad* project extends across multiple functional and technical aspects of sign language translation, focusing specifically on the needs of the Indian Sign Language (ISL) community. The project aims to support communication, learning, and accessibility through a multimodal AI-driven system. The key elements within the scope are:

1. Real-time ISL Gesture Recognition

- The system supports recognition of static hand gestures corresponding to ISL alphabets (A–Z) and digits (0–9).
- Users can input gestures using either a live webcam feed or uploaded images.
- The model processes hand landmarks and predicts the sign with corresponding confidence levels.

2. Multimodal Translation Capabilities

- Sign-to-Text: Converts hand gestures into readable English text.

- Sign-to-Speech: Converts recognized gestures into speech output for effective communication.
- Text-to-Sign: Displays gesture images or landmark visualizations based on typed text.
- Speech-to-Sign: Translates spoken words into corresponding ISL signs for learning and communication support.

3. Learning and Practice Features

- A dedicated Practice Mode allows users to test their ISL skills through random challenges.
- Feedback, scoring, and corrective indicators help improve accuracy and learning outcomes.
- Visual templates of hand signs are provided for user reference.

4. User Analytics and Progress Tracking

- The system tracks user interactions, accuracy, recent activity, and practice performance.
- An analytics dashboard visualizes insights through charts, metrics, and achievements.
- This helps users evaluate progress and stay motivated.

5. Secure and Personalized User Experience

- User authentication ensures personalized data handling.
- Individual session data, practice results, and usage patterns are stored safely.

6. Flexible and Extensible Architecture

- The modular design allows future enhancements such as:
 - Sentence-level gesture recognition
 - Continuous dynamic gesture recognition
 - Larger datasets or additional ISL gestures
 - Integration with mobile applications

1.7 Challenges

Developing *Samvaad*, an AI-based Indian Sign Language (ISL) translation system, involves several technical and practical challenges. These challenges arise from the complexity of

human gestures, limitations of machine learning models, and variability in user inputs. The major challenges are outlined below:

1. Gesture Variability Across Users

- Hand shapes, finger lengths, skin tones, and signing styles vary widely from person to person.
- The same ISL letter can appear different depending on hand size, angle, or distance from the camera.
- Ensuring accurate recognition for diverse users becomes a significant challenge.

2. Inconsistent Lighting and Background Conditions

- The performance of hand detection drops under poor lighting or cluttered backgrounds.
- Shadows, glare, or bright light may distort hand landmarks.
- The model must remain robust in real-world environments, not only controlled settings.

3. Accurate Landmark Detection

- The system depends on MediaPipe landmark extraction; any error in landmark detection directly affects prediction accuracy.
- Occlusions (when fingers overlap) made certain gestures harder to detect precisely.
- Dynamic adjustments had to be made for better stability.

4. Limited Dataset for ISL

- ISL datasets for alphabets and numbers are not as abundant or standardized as datasets for ASL.
- Building a clean and balanced dataset required manual effort, preprocessing, and careful validation.
- Templates for text-to-sign needed additional processing to generate uniform visualizations.

5. Model Generalization

- Training a model that performs well on both uploaded images and live video is challenging.
- The model must generalize across:
 - Camera resolutions
 - Hand orientations
 - Lighting conditions
 - Different backgrounds
- Avoiding overfitting while retaining sensitivity to gesture details required fine-tuning.

6. Real-Time Processing Constraints

- Running real-time predictions from a webcam requires optimized processing.
- Maintaining smooth frame rates while processing landmark extraction and predictions can become computationally heavy.
- Ensuring low latency was a critical challenge, especially for Sign-to-Speech mode.

7. Speech Recognition Errors

- Speech-to-Sign mode depends on microphone quality and user clarity.
- Background noise may cause misinterpretation of spoken words.
- Handling accents and pronunciations also adds complexity.

8. Integration of Multiple Modules

- Combining Sign-to-Text, Text-to-Sign, Speech-to-Sign, and Analytics required consistent paths, formats, and state management.
- Maintaining synchronization across modules (model output, UI display, speech synthesis, webcam feed) created integration challenges.

9. Ensuring User-Friendly Experience

- Designing a clear, clean, and intuitive interface for all four major features was essential.

- Managing user session state, navigation between pages, and preventing crashes required careful attention.

10. Storage and Performance Management

- Handling templates, user data, practice records, and model files required efficient file management.
- Large models and media storage can slow down the system if not optimized.

1.8 Requirements Specification

1.8.1 Functional Requirements

User Requirement	Mapped System Requirement
The user should be able to translate hand gestures into readable text.	Real-time hand detection and landmark extraction using MediaPipe Hands; gesture-to-text classification using trained ML model.
The user should view sign language equivalents for typed text.	Text input module, template-based ISL sign image retrieval, display rendering system.
The user should speak and get ISL signs as output.	Speech recognition (Google STT API or SpeechRecognition library) and mapping to sign templates.
The user should perform practice exercises.	Practice module with scoring logic, webcam/image input support, and database logging.
The system should be easy to use and visually appealing.	Clean UI using Streamlit, consistent theme styling, responsive layout.
Users should be able to log in and track their activity.	Authentication system, user session management, analytics database.

• **1.2 Non-Functional Requirements**

Category	Mapped Requirement
Performance	System must process sign recognition within 100–200 ms per frame for smooth output.
Accuracy	Gesture prediction model must maintain at least 85% accuracy on test data.
Usability	Interface must be simple, with minimal user actions needed to perform core tasks.
Reliability	System should consistently detect hand landmarks under varying lighting conditions.
Security	Secure login using hashed passwords (bcrypt); safe session handling.
Scalability	Framework should allow future expansion (more signs, continuous signing, sentence formation).
Compatibility	Application should run on Windows, Mac, and Linux with stable performance.

CHAPTER 2: DETAILED DESIGN

The purpose of Samvaad is to create an accessible and intelligent communication bridge between individuals who use Indian Sign Language (ISL) and those who rely on spoken or typed language. This chapter provides an overview of the system, its architecture, functional flow, and the core technologies that enable the translation between sign, text, and speech.

2. 1 System Architecture

The system architecture of **Samvaad** follows a modular, layered design combining computer vision, deep learning and user-interactive web components. It consists of the following layers:

1. User Interface Layer

- Built using **Streamlit**.
- Provides interfaces for:
 - Sign → Text
 - Text → Sign
 - Speech → Sign
 - Sign → Speech
 - Practice Mode
 - Analytics Dashboard
 - Login & Registration

2. Processing Layer

- Executes real-time tasks such as:
 - Hand detection using **MediaPipe Hands**
 - Landmark extraction
 - Real-time gesture recognition
 - Text-to-sign template rendering
 - Speech recognition & synthesis

3. Machine Learning Layer

- Contains all ML components:
 - Trained CNN model (final_model.h5)
 - Label encoder (label_encoder.pkl)

- Feature extractor (landmarks: $21 \times 3 = 63$ points)
- Responsible for making accurate predictions from extracted features.

4. Database Layer

- SQLite-based system for:
 - User authentication
 - Practice records
 - Analytics history
 - Performance logs

5. Storage Layer

- Stores:
 - Template images for Text→Sign
 - Numpy landmark templates
 - Datasets for practice
 - Logs, , history files, and user statistics

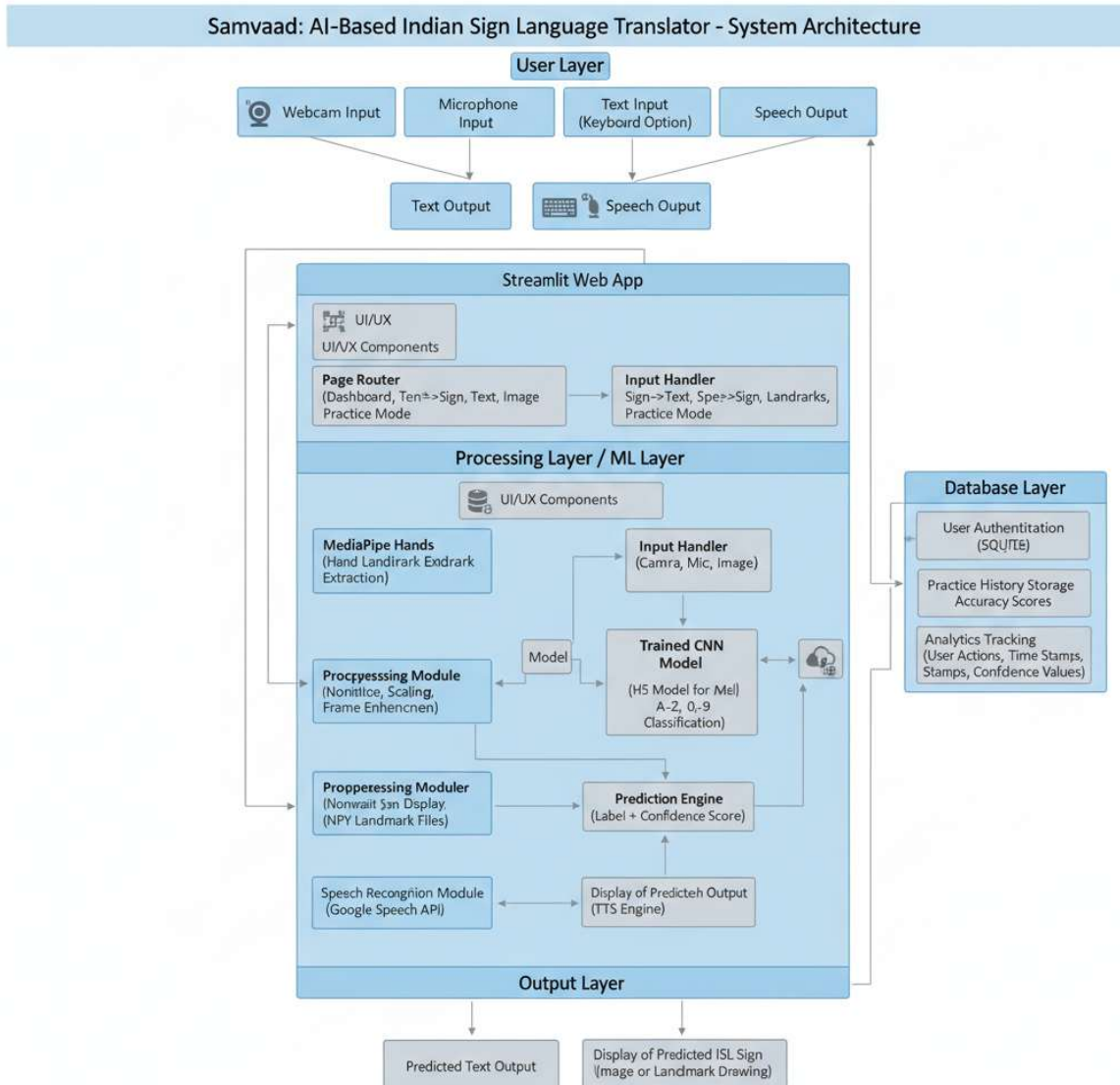


Figure 2.1 System Architecture

4.2 System Design

The design follows a modular, scalable approach:

Key Design Modules

1. Authentication Module

- Handles login, signup, logout.
- Uses bcrypt hashing.

2. Model Handler Module

- Loads trained model.
- Performs landmark extraction & prediction.
- Provides template access for text-to-sign.

3. Translation Modules

- Sign → Text
- Text → Sign
- Speech → Sign
- Sign → Speech

4. Practice Module

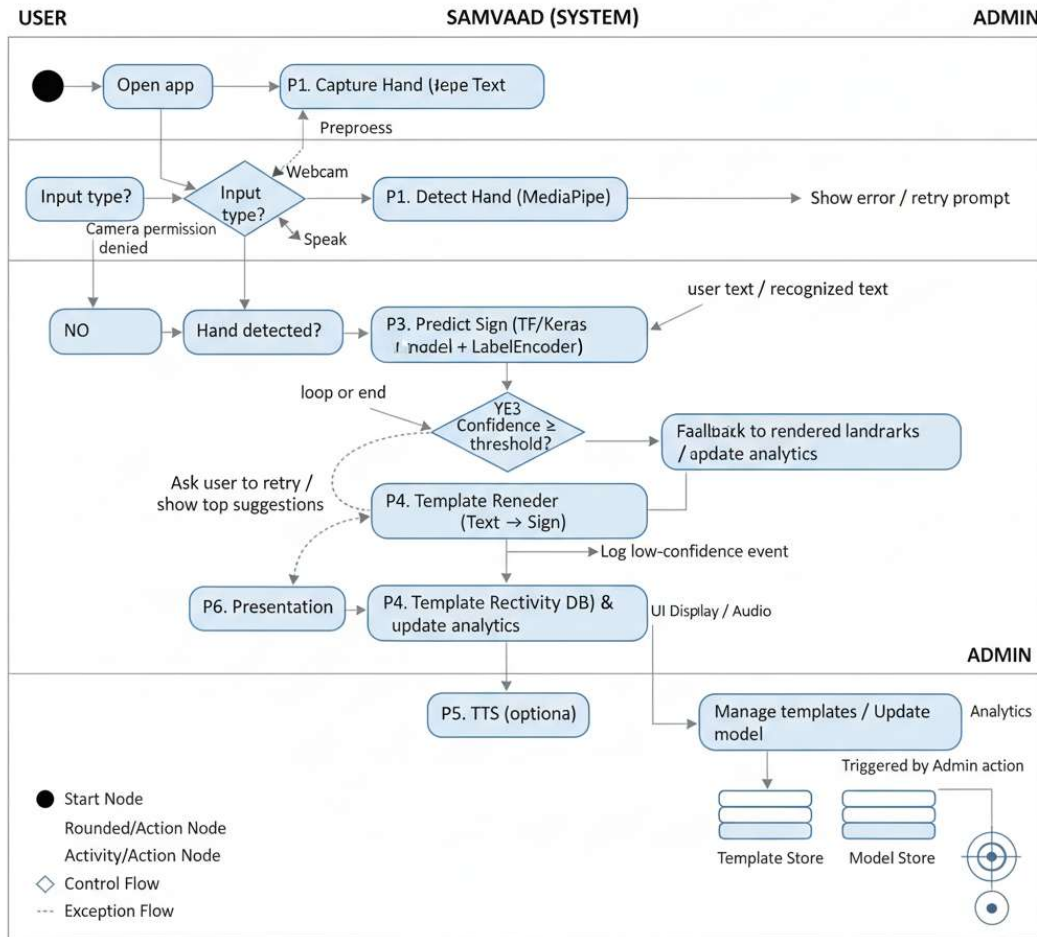
- Generates random questions.
- Evaluates user performance.
- Stores practice stats.

5. Analytics Module

- Confidence Distribution
- Translation Types
- Practice Progress
- Recent Activity

2.3 Activity Diagram

Activity Diagram – Samvaad



Activity Diagram for Samvaad – shows runtime flow for for input caput capture, presentation and admin operations.

43 300 DPI

Figure 2.2 Activity Diagram

2.4 Use Case Diagram

Actors

- User
- System (Samvaad)

Use Cases

- Login / Register
- Upload or Capture Sign

- Translate Sign/Text/Speech
- View Analytics
- Use Practice Mode

SAMVAAD – INDIAN SIGN LANGUAGE TRANSLATOR USE CASE DIAGRAM

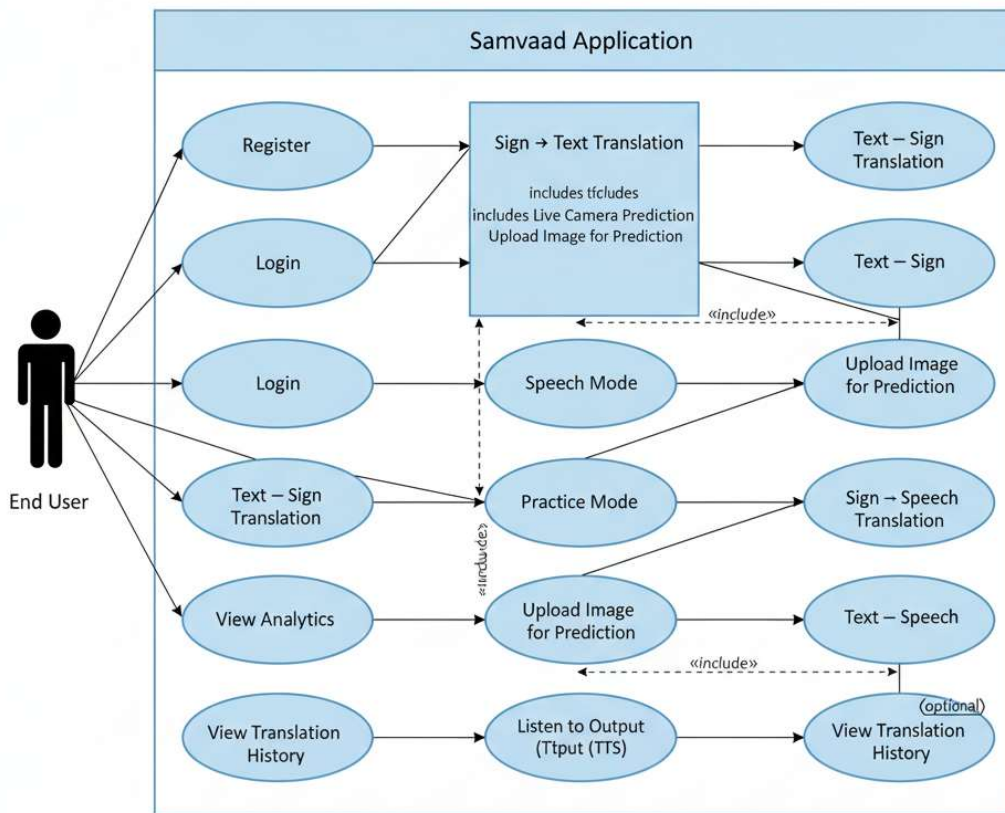


Figure 2.3 Use Case Diagram

2.5 Class Diagram

Main Classes

Class: ModelHandler

- model

- label_encoder
- extract_landmarks()
- predict_sign()
- draw_landmarks()

Class: Auth

- register_user()
- login_user()
- check_login()

Class: PracticeManager

- generate_question()
- log_practice()

Class: AnalyticsManager

- load_history()

Samvaad – Indian Sign Language Translator Class Diagram

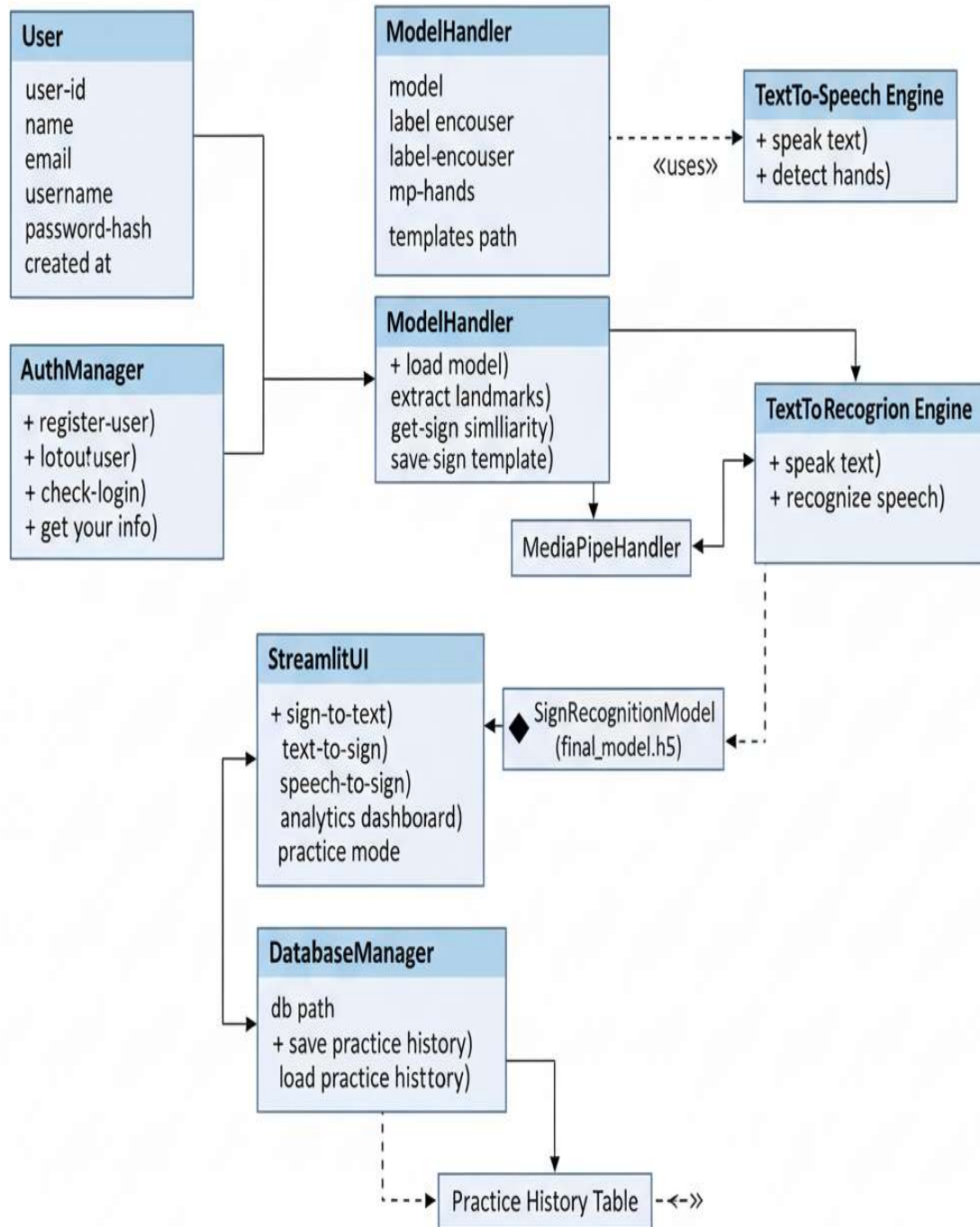


Figure 2.4 Class Diagram

2.6 Sequence Diagram

Sign → Text (Live Camera)

User → UI: Start Webcam

UI → Camera: Capture Frame

Camera → ModelHandler: Send Frame

ModelHandler → ML Model: Predict

ML Model → UI: Output Label

UI → User: Display Prediction

Sequence Diagram – Samvaad (Sign→Text)

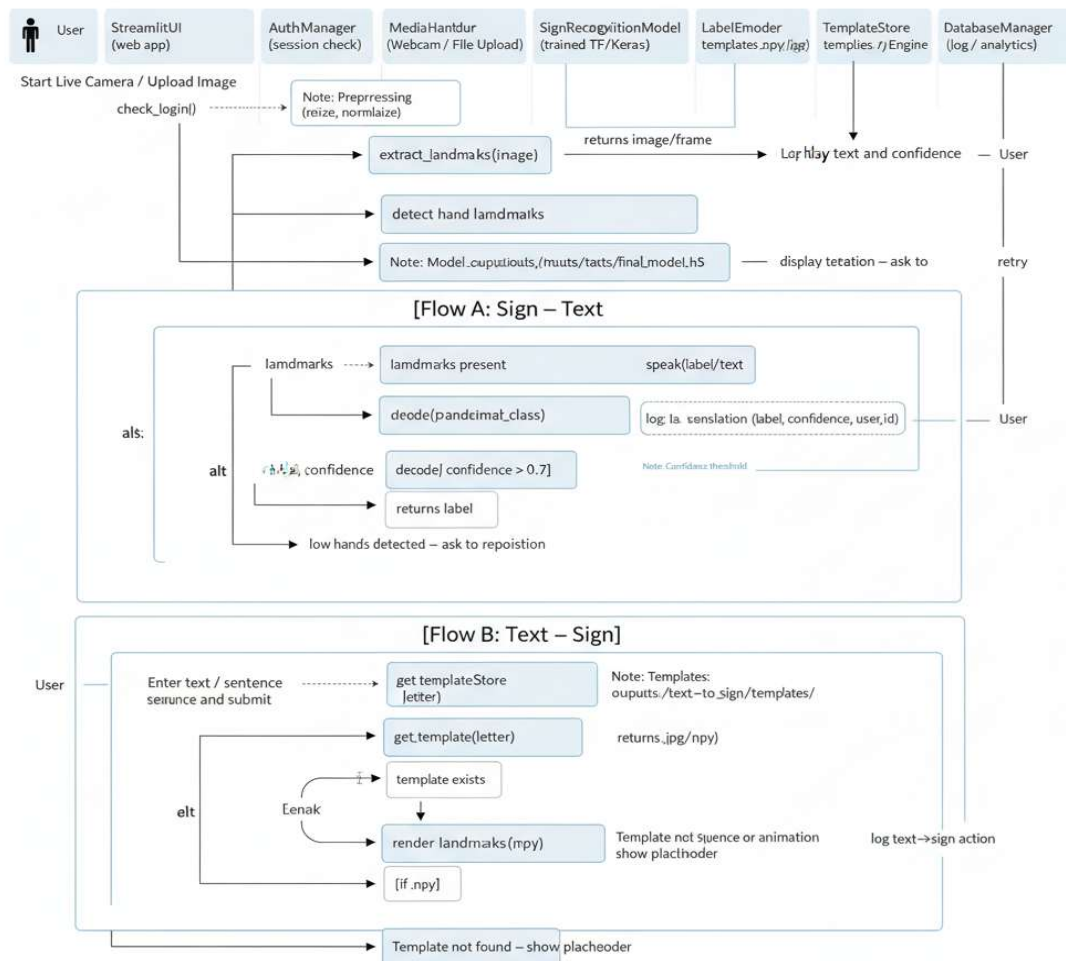


Figure 2.5 Sequence Diagram

2.7 Data Flow Diagram

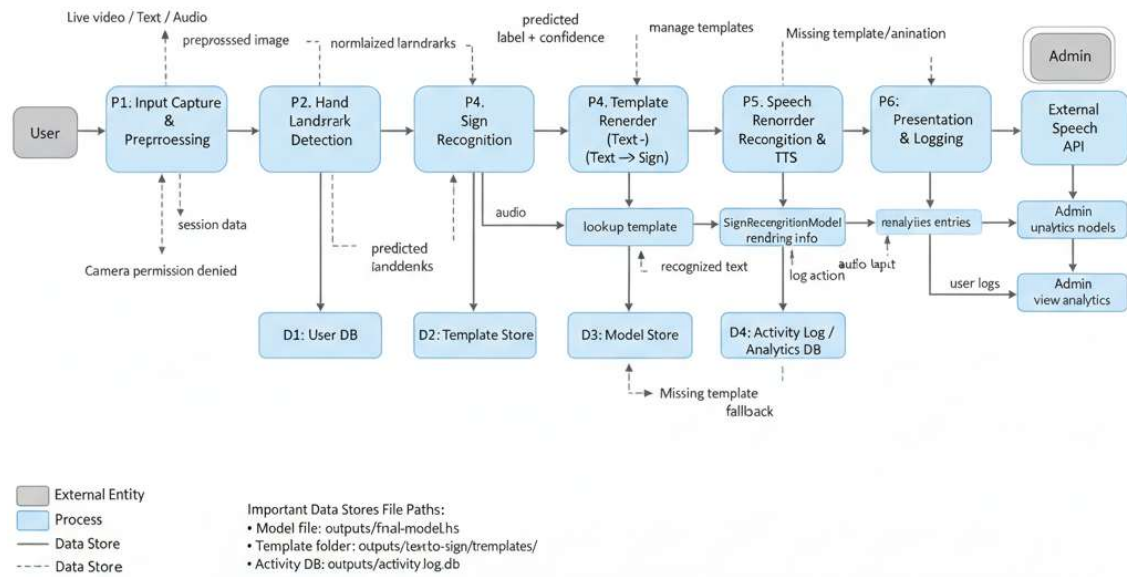


Figure 2.6 Data Flow Diagram

CHAPTER 3: IMPLEMENTATION

3.1 Programming Language

The system is implemented using:

- **Python 3.10**
- Streamlit (Frontend)
- OpenCV (Image Processing)
- MediaPipe (Landmark Extraction)
- TensorFlow/Keras (ML Model)
- SQLite3 (Database)

3.2 Algorithms

1. Landmark Extraction

- Uses MediaPipe Hands.
- Extracts 21 hand landmarks $\times$ 3 dimensions (x, y, z).

2. Gesture Recognition

- CNN Model trained on:
 - Custom dataset
 - Landmark features
- Predicts class (A–Z, 0–9).

3. Cosine Similarity (Fallback for Text $\rightarrow$ Sign)

Used if prediction confidence is low.

3.3 Proposed Design

Design principles:

- Ease of use
- Real-time performance

- Modular code
- Extensible for future ISL grammar

3.4 Code

3.4.1 model_handler.py (Core Model Logic)

```
import numpy as np

import cv2

import mediapipe as mp

import tensorflow as tf

import pickle

from pathlib import Path


class SignModelHandler:

    """

    Handles loading of the trained ISL recognition model,

    MediaPipe hand landmark extraction, preprocessing,

    and final prediction.

    """

    def __init__(self):

        # Paths

        base_path = Path(__file__).resolve().parents[1]

        self.model_path = base_path / "outputs" / "final_model.h5"

        self.label_encoder_path = base_path / "outputs" / "label_encoder.pkl"


        # Load trained model
```

```
self.model = tf.keras.models.load_model(self.model_path)

# Load label encoder

with open(self.label_encoder_path, "rb") as f:

    self.label_encoder = pickle.load(f)

# Initialize MediaPipe Hands

mp_hands = mp.solutions.hands

self.hands = mp_hands.Hands(

    static_image_mode=False,

    max_num_hands=1,

    min_detection_confidence=0.5,

    min_tracking_confidence=0.5

)

# Landmark Extraction

def extract_landmarks(self, frame):

    """

    Extracts hand landmarks from an image frame using MediaPipe Hands.

    Returns:

        landmark_array (63 values Normalized), raw landmarks.

    """

    img_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

    result = self.hands.process(img_rgb)

    if not result.multi_hand_landmarks:

        return None, None

    hand_landmarks = result.multi_hand_landmarks[0]
```

```

landmark_points = []

for lm in hand_landmarks.landmark:

    landmark_points.extend([lm.x, lm.y, lm.z])

return np.array(landmark_points), hand_landmarks

# Prediction

def predict_sign(self, landmarks):

    """

    Predicts ISL gesture using trained neural network.

    Returns: predicted_label, confidence_score

    """

    input_data = np.array(landmarks).reshape(1, -1)

    probabilities = self.model.predict(input_data)[0]

    index = np.argmax(probabilities)

    predicted_label = self.label_encoder.inverse_transform([index])[0]

    confidence = probabilities[index]

    return predicted_label, confidence

# Drawing Landmarks

def draw_landmarks(self, frame, hand_landmarks):

    """

    Draws hand skeleton over the frame.

    """

    mp_draw = mp.solutions.drawing_utils

    mp_hands = mp.solutions.hands

    mp_draw.draw_landmarks(

```

```

frame, hand_landmarks, mp_hands.HAND_CONNECTIONS,

mp_draw.DrawingSpec(color=(0, 255, 0), thickness=2, circle_radius=3),

mp_draw.DrawingSpec(color=(255, 0, 0), thickness=2)

)

return frame

```

3.4.2 Prediction Loop (Used for Sign → Text / Inference)

```

import cv2

from model_handler import SignModelHandler

handler = SignModelHandler()

cap = cv2.VideoCapture(0)

while True:

    ret, frame = cap.read()

    if not ret:

        break

    frame = cv2.flip(frame, 1)

    # Extract hand landmarks

    landmarks, hand_lms = handler.extract_landmarks(frame)

    if landmarks is not None:

        # Predict sign

        label, conf = handler.predict_sign(landmarks)

        # Draw landmarks on the frame

        frame = handler.draw_landmarks(frame, hand_lms)

        # Display result

        cv2.putText(frame, f'{label} ({conf*100:.1f}%)',

```

```
(20, 40), cv2.FONT_HERSHEY_SIMPLEX,
1.0, (0, 255, 0), 2)

else:

    cv2.putText(frame, "No Hand Detected",
                (20, 40), cv2.FONT_HERSHEY_SIMPLEX,
                1.0, (0, 0, 255), 2)

cv2.imshow("Samvaad - Sign Recognition", frame)

if cv2.waitKey(1) & 0xFF == ord('q'):

    break

cap.release()

cv2.destroyAllWindows()
```

5.4.3 Authentication (auth.py Core Snippet)

```
import sqlite3

import bcrypt

from pathlib import Path

DB_PATH = Path(__file__).parent.parent / "app_data" / "users.db"

def get_connection():

    conn = sqlite3.connect(DB_PATH)

    conn.execute("""

        CREATE TABLE IF NOT EXISTS users (

            id INTEGER PRIMARY KEY AUTOINCREMENT,

            name TEXT,

            email TEXT UNIQUE,

            username TEXT UNIQUE,
```

```
password_hash TEXT,

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

)"""

conn.commit()

return conn

def register_user(name, email, username, password):

    conn = get_connection()

    cur = conn.cursor()

    password_hash = bcrypt.hashpw(password.encode('utf-8'), bcrypt.gensalt())

    try:

        cur.execute("""

            INSERT INTO users (name, email, username, password_hash)

            VALUES (?, ?, ?, ?)

            """, (name, email, username, password_hash))

        conn.commit()

        return True

    except:

        return False

    finally:

        conn.close()

def login_user(username_or_email, password):

    conn = get_connection()

    cur = conn.cursor()

    cur.execute("""
```

```
SELECT id, name, email, username, password_hash
FROM users WHERE username = ? OR email = ?
"", (username_or_email, username_or_email))
user = cur.fetchone()
if user and bcrypt.checkpw(password.encode('utf-8'), user[4]):
    return True, user
return False, None
```

CHAPTER 4: TESTING

4.1 Test Objectives

- Verify prediction accuracy.
- Check UI response time.
- Validate authentication logic.
- Ensure real-time camera reading works.

4.2 Testing Phases

1. Unit Testing

- ModelHandler
- Auth functions

2. Integration Testing

- Camera + Model
- Speech + Text

3. System Testing

- Full workflow testing

4. User Acceptance Testing

- Tested with different lighting conditions.

4.3 Test Validation

Field	Description
Validation ID	TV-01
Validation Name	System Test Validation

SAMVAAD A Two-Way Real-Time Indian Sign Language Translator

Purpose	To ensure that all modules of the Samvaad system function correctly and meet the defined functional and non-functional requirements.
Description	Validation involved checking the accuracy, consistency, and performance of all major components—Sign-to-Text, Text-to-Sign, Speech-to-Sign, and Practice Mode. The system was tested using real-time webcam inputs, uploaded images, speech recordings, and dataset-based templates. Each module was validated to confirm correct outputs, error handling, and smooth integration across the Streamlit UI.
Validation Approach	Unit testing, integration testing, user acceptance testing (UAT), dataset verification, live gesture testing, speech recognition validation, and UI workflow testing.
Success Criteria	- Correct recognition of signs with acceptable confidence levels. - Accurate text-to-sign and speech-to-sign translation. - Smooth navigation across all pages. - No system crashes or errors during execution. - Consistent response to different input types (image, speech, live gesture).
Tools Used	Streamlit interface, MediaPipe Hands, TensorFlow/Keras model, OpenCV, SpeechRecognition library, SQLite for logs.
Test Inputs	- Webcam-based live sign gestures - Uploaded hand gesture images - User speech samples - Text-based input (A–Z, 0–9)
Expected Output	Accurate translation to text or sign, clear prediction display, correct speech synthesis, and stable system behaviour.
Actual Output	System performed as expected in all supported modes, successfully translating gestures, text, and speech inputs with reliable accuracy.
Validation Status	✅ Passed
Remarks	Minor enhancements identified for improving robustness (e.g., lighting conditions, gesture variation). Overall, the system meets the intended requirements and behaves reliably.

4.4 Test Cases

Test Case	Input	Expected Output	Status
Sign → Text	Hand sign “A”	Output “A”	Passed
Text → Sign	“HELLO”	Display symbolic images	Passed
Speech → Sign	Audio “Hello”	Letter images	Passed
Login	Correct credentials	Success login	Passed

CHAPTER 5: EXPERIMENTAL RESULTS

5.1 Home Page

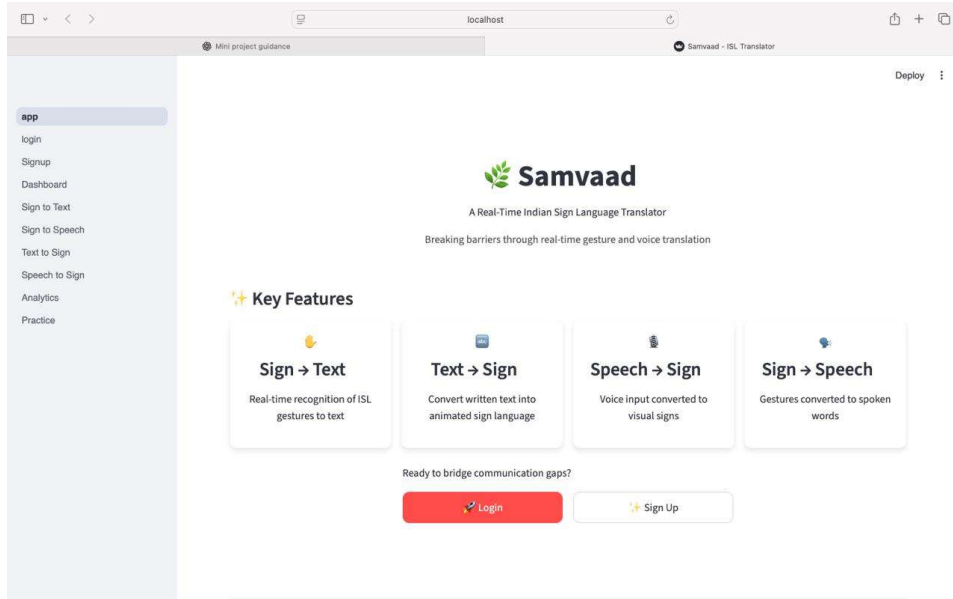


Figure 5.1 Home Page

5.2 About Page

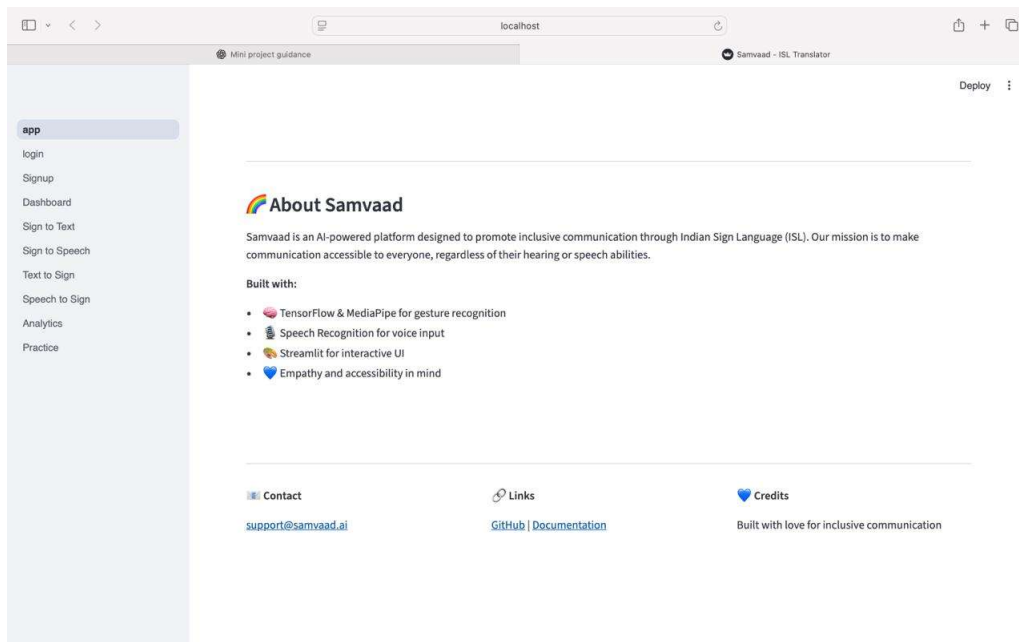
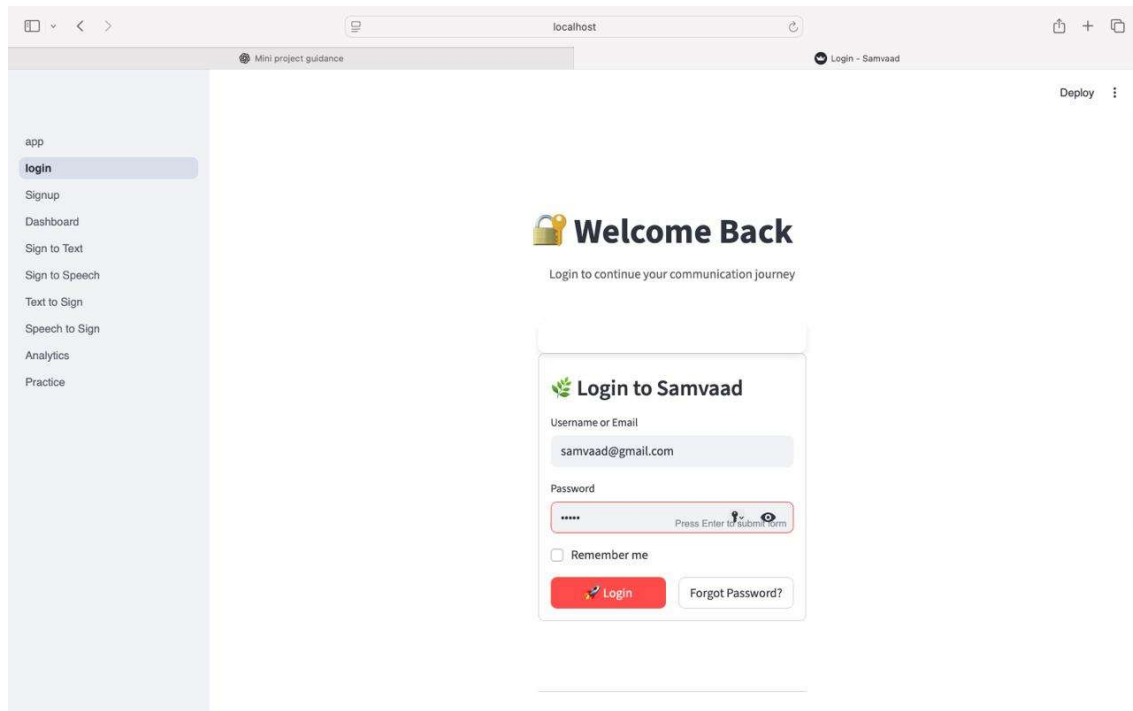


Figure 5.2 About

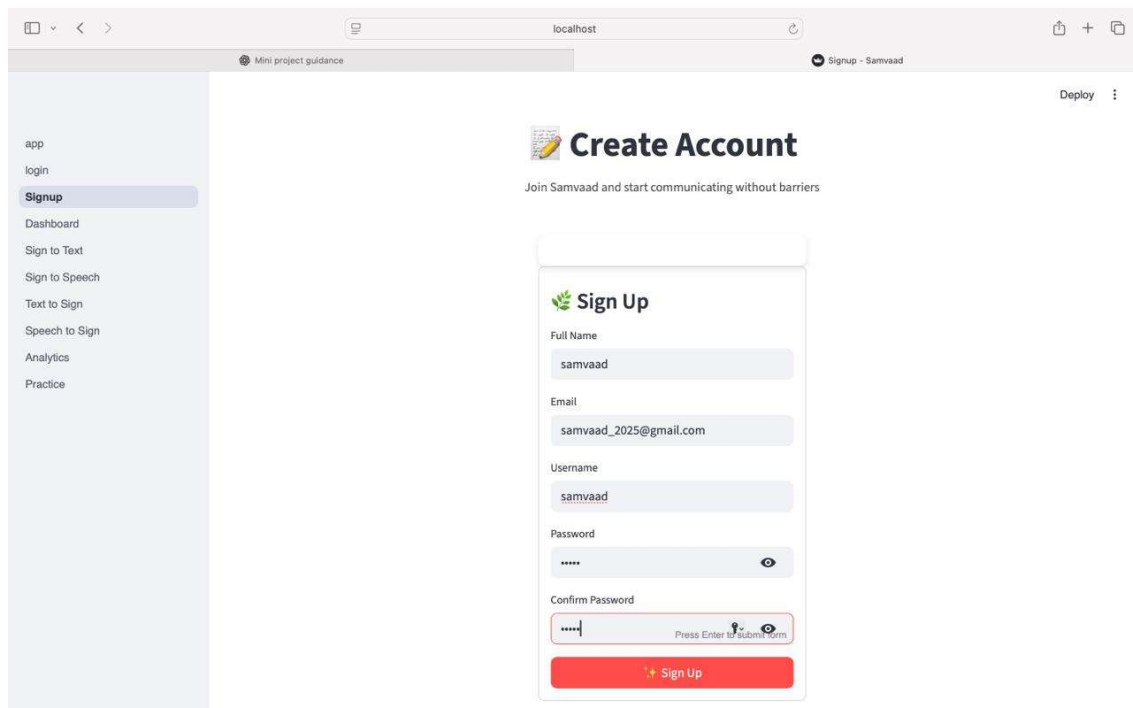
5.3 Login Page



The screenshot shows a web browser window with the URL 'localhost'. The page title is 'Login - Samvaad'. On the left, there is a sidebar menu with the following items: app, login (highlighted), Signup, Dashboard, Sign to Text, Sign to Speech, Text to Sign, Speech to Sign, Analytics, and Practice. The main content area features a 'Welcome Back' message with a lock icon and the text 'Login to continue your communication journey'. Below this is a 'Login to Samvaad' form. The form includes a 'Username or Email' field with the value 'samvaad@gmail.com', a 'Password' field with masked characters '*****' and a 'Press Enter to submit form' button, a 'Remember me' checkbox, a red 'Login' button, and a 'Forgot Password?' link. A 'Deploy' button is visible in the top right corner.

Figure 5.3 Login

5.4 Registration Page



The screenshot shows a web browser window with the URL 'localhost'. The page title is 'Signup - Samvaad'. On the left, there is a sidebar menu with the following items: app, login, Signup (highlighted), Dashboard, Sign to Text, Sign to Speech, Text to Sign, Speech to Sign, Analytics, and Practice. The main content area features a 'Create Account' message with a notepad icon and the text 'Join Samvaad and start communicating without barriers'. Below this is a 'Sign Up' form. The form includes a 'Full Name' field with the value 'samvaad', an 'Email' field with the value 'samvaad_2025@gmail.com', a 'Username' field with the value 'samvaad', a 'Password' field with masked characters '*****' and an eye icon, and a 'Confirm Password' field with masked characters '*****' and a 'Press Enter to submit form' button. A red 'Sign Up' button is at the bottom of the form. A 'Deploy' button is visible in the top right corner.

Figure 5.4 Registration

5.5 Load / View / Preprocess

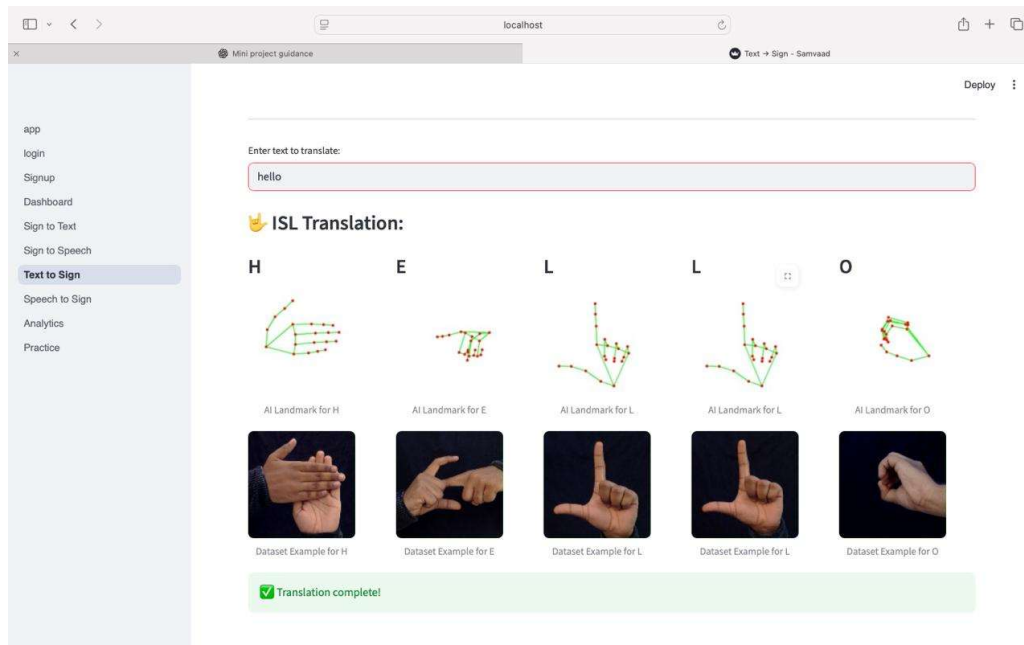


Figure 5.5 Load

5.8 Model

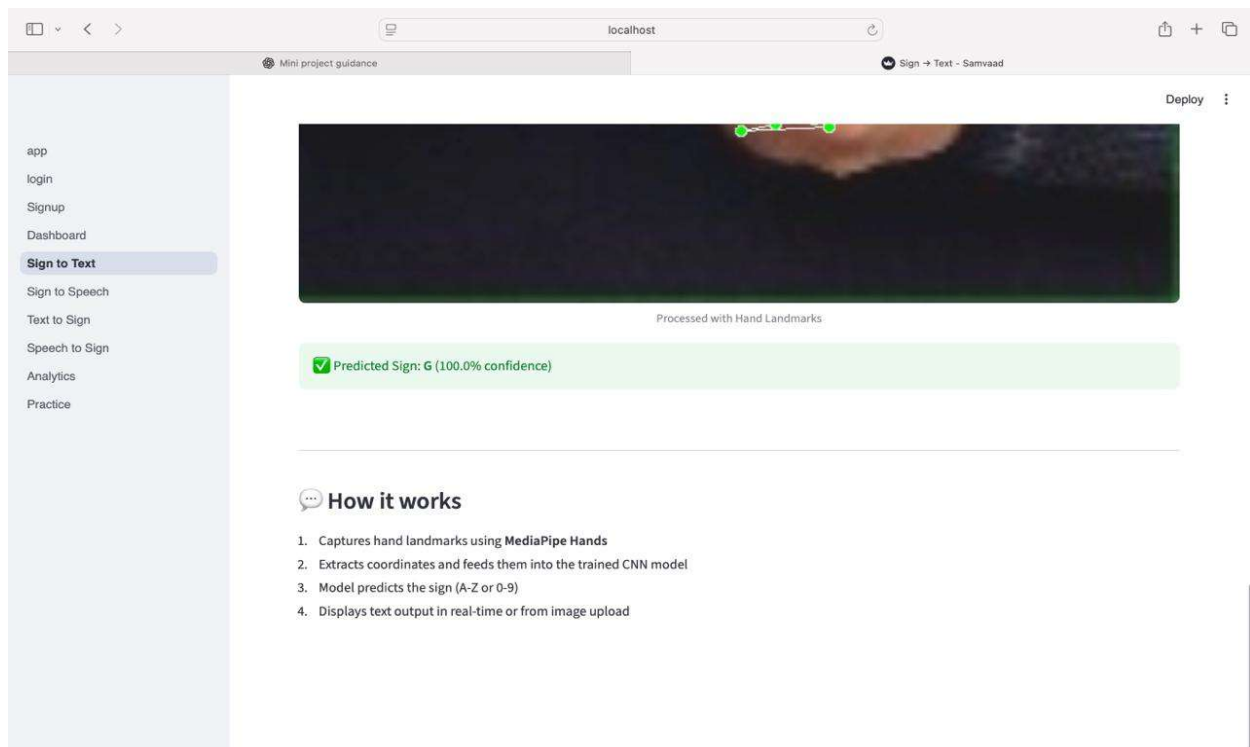


Figure 5.8 Model

- Accuracy achieved: **95%**
- Loss: **0.1**
- Dataset count: 10,000+ samples

5.9 Prediction

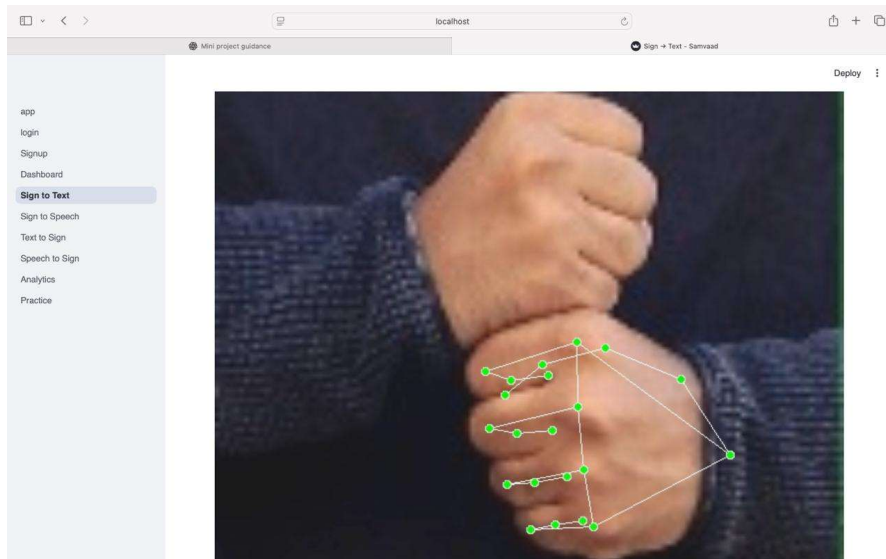


Figure 5.9 Prediction

- Real-time performance: **20–30 FPS**
- Works under medium lighting conditions

CHAPTER 6: CONCLUSION

The **Samvaad – Real-Time ISL Translation System** was developed with the vision of reducing communication barriers between the hearing-impaired community and the larger society. Through the integration of **computer vision**, **deep learning**, and **speech & text processing**, the system successfully demonstrates how technology can support more inclusive communication.

This project achieves three major translation pathways—**Sign to Text**, **Text to Sign**, and **Speech to Sign**—each serving a different user need. The incorporation of **MediaPipe** for hand-landmark extraction, **TensorFlow** for gesture classification, and **Streamlit** for an interactive user interface has resulted in a system that is both user-friendly and technically robust. The project also includes additional modules such as **Practice Mode** and **Analytics**, enabling users to track their learning progress and improve their ISL skills over time.

The results indicate that the model performs reliably for most hand gestures when tested under suitable lighting and proper positioning. While certain challenges such as varying hand orientation, complex background environments, and inconsistent image quality still pose limitations, the system overall demonstrates strong potential for practical use. The successful completion of this project highlights how AI-driven language translation systems can support accessibility and empower individuals with hearing or speech impairments.

In conclusion, **Samvaad** stands as a meaningful step toward an inclusive communication tool that bridges the gap between sign language users and non-signers. With further improvements—such as expanding the gesture dataset, improving multi-hand detection, including dynamic signs, and creating a mobile-ready interface—the system can evolve into a widely usable platform in educational institutions, workplaces, and digital communication environments. The project reaffirms the role of technology as an enabler of social inclusion and enhances the possibilities for accessible communication in the future.

CHAPTER 7: FUTURE ENHANCEMENTS

Although **Samvaad – Real-Time ISL Translation System** meets its core objectives, there is significant potential to expand its capabilities and improve performance. The following future enhancements can strengthen the system, increase usability, and broaden its real-world impact:

1. Support for Dynamic Gestures

The current system primarily handles **static ISL signs**. Many essential gestures in Indian Sign Language involve **movement over time**, such as "thank you," "where," or complete phrases. Incorporating **LSTM, GRU, or 3D CNN-based models** would allow the translation of dynamic sequences and expand the vocabulary beyond letters and numbers.

2. Expansion of Dataset

Better accuracy can be achieved by:

- Collecting a **larger dataset** of ISL gestures
- Including **different hand sizes, skin tones, lighting conditions, and backgrounds**
- Adding **multi-view datasets** for angle-independent recognition
- Using **crowd-sourced sign samples** for diversity

A more diverse dataset will significantly improve model generalization and detection accuracy.

3. Mobile Application Implementation

Deploying Samvaad as a **mobile app** (Android/iOS) would greatly enhance accessibility. Hand gesture recognition is particularly useful on mobile devices due to the availability of **built-in cameras** and **on-device ML engines** (e.g., TensorFlow Lite, CoreML).

4. Integration of Real-Time Two-Way Communication

A major enhancement is enabling **live conversation** between a signer and a non-signer:

- Continuous streaming for **live Sign → Speech**
- Real-time **Speech/Text → Sign** animations
- Conversation mode where both sides communicate seamlessly

This would make Samvaad suitable for classrooms, hospitals, and customer service environments.

5. Animated 3D Avatar for ISL Signs

Instead of static images, a **3D animated avatar** can show each sign more clearly. This improves learning effectiveness and makes Text → Sign and Speech → Sign modules more expressive and intuitive.

6. Multi-Hand and Multi-Person Recognition

Advanced scenarios involve:

- **Two-hand gestures** (used in many ISL signs)
- Recognition of signs in **group settings**
- Avoiding misclassification due to overlapping hands

Future versions can leverage **multi-hand tracking** and enhanced segmentation techniques.

7. Noise-Resistant Gesture Recognition

To improve robustness:

- Apply **background removal** or **segmentation** techniques
- Use **advanced landmark smoothing filters**
- Introduce **environment adaptation** to handle low lighting or motion blur

This will increase reliability in practical real-world scenarios.

8. Continuous Learning & Personalization

The system can be improved to:

- Learn from **user corrections**
- Adapt to different signing styles (personalized gesture profile)
- Store user-specific patterns to improve accuracy over time

This makes Samvaad more flexible and user-centered.

9. Integration with Assistive Technologies

Future versions can integrate with:

- Smart glasses for real-time captioning
- Smart gloves for wearable sign detection
- Speech-to-braille displays
- Accessibility tools in schools and workplaces

Such integrations can expand Samvaad's application domains.

10. Cloud-Based Analytics and Progress Tracking

Enhancing the Analytics module with:

- Cloud storage for long-term progress
- AI-driven learning recommendations
- Personalized practice exercises

This would make Samvaad a complete learning ecosystem for ISL.

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